

Name: _____

Date: _____

Class: _____

Element J: Documentation of Evaluation

Stakeholder Evaluation #1

Name of evaluator: Mr. O'Farrell

Justification for how this evaluator is a stakeholder: Teaches TV Production at Walpole High School and worked for Fox News as a television production technician.

GREAT PRESENTATION

Walpole Engineers Design-a-Thon - Judge's Feedback

Judge: Mr. O'Farrell Team/Project Title: 2 Rampage

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
Problem definition and justification are clear and detailed.	Problem definition and justification are somewhat clear and objective.	Problem definition and justification are stated but unclear.	3
Design requirements are consistently objective and measurable and highly likely to lead to a viable solution.	Design requirements are generally objective and measurable.	Design requirements are insufficiently measurable and have little possibility of leading to a viable solution.	3
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			
The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is consistently clear, detailed, and supported by relevant data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is generally clear and contains some detail and relevant supporting data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is vague and is missing relevant details and/or relevant supporting data.	3 SHOULD ALL STEPS IN DEVELOPMENT
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
The proposed solution is well substantiated with STEM content, principles and practices.	The proposed solution is partially substantiated with STEM content, principles and practices.	The proposed solution is lacking substantiation with STEM content, principles and practices, though an attempt was made.	3 WELL DOCUMENTED STEPS
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas you came up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts?			

- Do you have any pictures or physical models of your previous or early attempts? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version?	The design solution chosen is well justified and satisfies all design requirements. Strengths and weaknesses are comprehensively noted and explained well.	Strengths and weaknesses of the design solution are noted and explained.	The discussion lacks analysis related to strengths and weaknesses of the solution in meeting the design requirements.	3	
Learning Goal: Create a prototype. - How did you make your current prototype? - What challenges did you face, and how did you overcome them? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version? - Were there any limitations or constraints that impacted what you were able to build?					
The prototype is constructed with enough detail to assure that objective data on all or nearly all design requirements could be determined.		The prototype is constructed with enough detail to assure that objective data on some design requirements could be determined.		The prototype is not constructed with enough detail to assure that objective data on at least one design requirement could be determined.	3 LOTS OF GREAT IDEAS
Learning Goal: Create and implement a Test Plan to evaluate the performance of the design solution. - What have you learned or what are you hoping to learn from testing your prototype today? - Who is your intended user and how did you consider their input in your Test Plan? - Can you measure any quantifiable constraints from your Design Brief? If so, what is an acceptable measure or result?					
The prototype testing plan addresses all or nearly all of the high priority design requirements.		The prototype testing plan addresses many of the high priority design requirements.		The prototype testing plan addresses a few of the high priority design requirements.	3
Learning Goal: Use various engineering communication methods.					
Student fully communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts.		Student partially communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts.		Student minimally communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts.	3
Student is fully able to effectively communicate important details of an object through sketching.		Student is partially able to effectively communicate important details of an object through sketching.		Student is minimally able to effectively communicate important details of an object through sketching.	3
Judge's Comments: <u>WELL PREPARED - GREAT DEMO SKETCHES</u> <u>EXCELLENT COMMUNICATION w/ ME AS THEY DESIGNED</u>					



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Stakeholder Evaluation #2

Name of evaluator: Ms. Reuther

Justification for how this evaluator is a stakeholder: Teaches Engineering Design

Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.

- Transcript: "Kyle- Intro= Very good. Sean- the process. Matt- Testing. Add balance to it and not top heavy"

Stakeholder Evaluation #3

Name of evaluator: Professor Love

Justification for how this evaluator is a stakeholder: Engineering Professor at Northeastern, and has 8 years of experience working as a shoe designer for Reebok.

Walpole Engineers Design-a-Thon - Judge's Feedback

Judge: Prof. Love Team/Project Title: Team Nanpage - Sean, Kyle, Matt, Mic Hecker

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
Problem definition and justification are clear and detailed.	Problem definition and justification are somewhat clear and objective.	Problem definition and justification are stated but unclear.	3
Design requirements are consistently objective and measurable and highly likely to lead to a viable solution.	Design requirements are generally objective and measurable.	Design requirements are insufficiently measurable and have little possibility of leading to a viable solution.	3
Learning Goal: Plan and conduct research by gathering relevant and credible data, facts, and information. - What evidence is there that your current design is based on your observations and interview of your primary stakeholder? - Where did you draw inspiration from for your current design? - What has your primary stakeholder already tried? How did you apply this information to your current design?			
The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is consistently clear, detailed, and supported by relevant data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is generally clear and contains some detail and relevant supporting data.	The analysis of past and current attempts to solve the problem — including both strengths and shortcomings — is vague and is missing relevant details and/or relevant supporting data.	3
Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
The proposed solution is well substantiated with STEM content, principles and practices.	The proposed solution is partially substantiated with STEM content, principles and practices.	The proposed solution is lacking substantiation with STEM content, principles and practices, though an attempt was made.	3
Learning Goal: Evaluate solution alternatives and select a final design by considering assumptions, trade offs, criteria, and constraints. - What other ideas you came up with during the brainstorming phase? - How did you make the decision to focus on your current design? - Can you walk through all the different phases/versions that your current design/prototype went through? - What did you learn or take away from all the previous versions or attempts?			



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- Do you have any pictures or physical models of your previous or early attempts? - What do you really like about your current design/prototype? - What are you looking forward to improving in the next version?			
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Student fully communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	Student partially communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	Student minimally communicates key aspects of a design through the creation of written documents and/or presentation that addresses all prompts	3?
Student is fully able to effectively communicate important details of an object through sketching	Student is partially able to effectively communicate important details of an object through sketching	Student is minimally able to effectively communicate important details of an object through sketching	3?
Judge's Comments: <i>good presentation, shared speaking roles, covered the rubric, I haven't seen your documentation but slide looked great! Great work! Prototypes really helped you iterate effectively!</i>			

Field Expert Evaluation #1

Name of evaluator: Mr. O'Farrell

Justification for how this evaluator is a demonstrably-qualified field expert (professionals who spend at least a portion of their time working in a field directly related to your project): Teaches TV Production at Walpole High School and worked for Fox News as a television production technician.

Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.



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GREAT PRESENTATION

Walpole Engineers Design-a-Thon - Judge's Feedback

Judge: Mr. O Team/Project Title: 2 Rampage

Proficient (3)	Developing (2)	Novice (1)	Score
Learning Goal: Identify and describe a problem that can be solved with a potentially new product or process. - Can you explain what the problem is and the process you used to come to that conclusion? - What kind of input/feedback did your primary stakeholder provide for the problem and/or design requirements? - How did empathy play a role in determining the problem and design requirements? - How did you decide which design requirement was the most important to focus on?			
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Learning Goal: Articulate appropriate STEM practices and principles in the design. What kind of research did you do to learn more about the problem and how did you apply that to your current design?			
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Student is fully able to effectively communicate important details of an object through sketching.	Student is partially able to effectively communicate important details of an object through sketching.	Student is minimally able to effectively communicate important details of an object through sketching.	3	3
Judge's Comments: <u>WELL PREPARED - GREAT DEMO SPEECHES</u> <u>EXCELLENT COMMUNICATION W/ ME AS THEY DESIGNED</u>				

Field Expert Evaluation #2

Name of evaluator: Mr. Alan

Justification for how this evaluator is a demonstrably-qualified field expert (professionals who spend at least a portion of their time working in a field directly related to your project): Teaches Film class at WHS, lead mentor for the WHS film festival which uses camera and microphone equipment



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Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.



Alan, Michael
to me ▾

May 8, 2024, 9:58 AM (5 days ago) ☆ ↶ ⋮

Very cool! My concern with this product is that the cover would hinder the input of sound. Have you tried it compared with a wireless microphone that does not have the cover to hear if there is any difference?



Alan, Michael
to me ▾

May 8, 2024, 11:52 AM (5 days ago)

Is this device to protect the microphone should it fall?

Field Expert Evaluation #3

Name of evaluator: Jack

Justification for how this evaluator is a demonstrably-qualified field expert (professionals who spend at least a portion of their time working in a field directly related to your project): TV Production student at WHS who uses camera and microphone products regularly during class, including mic holders

Documentation of the project evaluation, which could include copies of emails, transcripts of conversations, filled out feedback forms, etc.

Mic Holder Feedback > Inbox x

 Spillane, Jack
to me ▾

12:32 PM (0 minutes ago) ☆ ↶ ⋮

Hello Team Rampage, thanks for sharing the mic holder design with me. I think the 3d printed mic holder will aid the TV production studio more than previous designs. I like how you added the top half of the head of the design to keep the microphone from falling out.

Jack Spillane

This email is sent from a Walpole Public Schools student account. Forward any inappropriate messages to mailadmin@walpole.k12.ma.us

Synthesis of the Project External Evaluation

Summarize all of the evaluators' comments in a way that highlights meaningful trends with specifics mentioned. Answer the questions, "What does the external evaluation mean?" and "What are you taking away from the external evaluation?"

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External evaluation is feedback from people who did not participate in the making of the prototype. We got feedback from stakeholders who were affected by the making of the project, and field experts who were able to provide educated feedback on the quality of the prototype. All the stakeholders and field experts were very satisfied with the mic holder. Some people were worried about the strength of the prototype protecting the microphone, but we assured them they would be strong enough using the data from the testing plan.